

SECOND SEMESTER (CBCSS-UG) DEGREE EXAMINATION, APRIL 2021**Mathematics****MTS 2B 02—CALCULUS OF SINGLE VARIABLE – I****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 2B 02—CALCULUS OF SINGLE VARIABLE – I

(Multiple Choice Questions for SDE Candidates)

1. The domain of $f(x) = \sqrt{x^2 - 1}$ is :

(A) $[1, \infty)$.

(B) $(-\infty, -1] \cup [1, \infty)$.

(C) $(-\infty, \infty)$.

(D) $(0, \infty)$.

2. $\lim_{x \rightarrow 0} \frac{\tan(x)}{x}$ is :

(A) 0.

(B) 1.

(C) π .

(D) Limit does not exist.

3. Which of the following functions are continuous for all real number x ?

(A) $\tan(x)$.

(B) $\frac{1}{x}$.

(C) $\sec(x)$.

(D) $\exp x$.

4. $\lim_{x \rightarrow a} f(x) = L$ if and only if :

(A) $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} f(x) = L$.

(B) $f(a) = L$.

(C) $\lim_{x \rightarrow a} f(x) = L$.

(D) None of the above.

5. On $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $f(x) = \cos(x)$ takes on :

(A) A maximum value of 1 (once) and a minimum value of 0 (twice).

(B) A maximum value of 1 (once) and no minimum value.

(C) A minimum value of 0 (twice) and no maximum value.

(D) A maximum value of 1 (once) and a minimum value of 0 (once).

6. Consider the function :

$$f(x) = \begin{cases} x+1 & -1 < x < 0 \\ 0 & x = 0 \\ x-1 & 0 < x < 1. \end{cases}$$

Then which of the following statements is NOT true ?

- (A) f is continuous at every point of $[-1, 1]$, except at $x = 0$.
- (B) f has a non-removable discontinuity at $x = 0$.
- (C) f has neither a highest nor a lowest point on $[-1, 1]$.
- (D) f has the highest value 1 and the lowest value -1 on $[-1, 1]$.

7. What are the critical points of f when $f'(x) = (x-1)(x-2)$?

- (A) 0, 1 and 2.
- (B) -1 and -2 .
- (C) 1 and 2.
- (D) No critical points.

8. The value or values of c that satisfy the equation $\frac{f(b)-f(a)}{b-a} = f'(c)$ in the conclusion of Mean

Value Theorem for the function $f(x) = x^2 + 2x - 1$ and the interval $[0, 1]$ is :

- (A) 1.
- (B) $\frac{1}{2}$.
- (C) $\frac{1}{3}$.
- (D) $\frac{1}{4}$.

9. The tangent at the point of inflection is called _____.

- (A) Inflectional tangent.
- (B) Vertical tangent.
- (C) Asymptote.
- (D) None of these.

10. $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}$:

(A) $\frac{5}{3}$.

(B) 5.

(C) 0.

(D) ∞ .

11. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(4x)}$:

(A) 1.

(B) $\frac{4}{3}$.

(C) $\frac{3}{4}$.

(D) 0.

12. $\lim_{x \rightarrow 0} \frac{|x|}{x}$:

(A) 1.

(B) 0.

(C) Does not exist.

(D) -1.

13. Find the maximum value of $f(x) = x - 2\ln x$ on the interval $[1, 4]$:

(A) 1.

(B) $3 - 2\ln 3$.

(C) $2 - 2\ln 2$.

(D) 2.

14. The linearization of a function $f(x) = \cos(2x)$ at the $x = \frac{1}{2}$.

(A) $y = \cos(1) - 2\sin(1)\left(x - \frac{1}{2}\right)$.

(B) $y = \cos(2) - 2\sin(2)\left(x - \frac{1}{2}\right)$.

(C) $y = \frac{1}{2} - 2\sin(1)\left(x - \frac{1}{2}\right)$.

(D) $y = \cos(1) + \sin(1)\left(x - \frac{1}{2}\right)$.

15. If $y = 12\sqrt{x^3} + 8$, then $\frac{dy}{dx}$ is :

(A) $18x$.

(B) $18\sqrt{x^5}$.

(C) $18x$.

(D) $18\sqrt{x}$.

16. If $y = \sin 3x$, then dy is _____.

(A) $dy = (\cos 3x) dx$.

(B) $dy = (-3 \cos 3x) dx$.

(C) $dy = (3 \cos 3x) dx$.

(D) $dy = (3 \sin 3x) dx$.

17. $d\left(\frac{x}{x+1}\right) =$ _____.

(A) $\frac{dx}{x+1}$.

(B) $\frac{x dx}{x+1}$.

(C) $\frac{x dx}{(x+1)^2}$.

(D) $\frac{dx}{(x+1)^2}$.

18. The radius r of a circle increases from $r_0 = 10$ m to 10.1 m. Estimate the increase in the circle's area A by calculating dA .

(A) $dA = 2\pi m^2$.

(B) $dA = -2\pi m^2$.

(C) $dA = \pi m^2$.

(D) $dA = -\pi m^2$.

19. Define norm of a partition :

(A) The norm of a partition P is the first subinterval length.

(B) The norm of a partition P is the average of partition's subinterval length.

(C) The norm of a partition P is the partition's shortest subinterval length.

(D) The norm of a partition P is the partition's longest subinterval length.

20. If $f(x) \geq g(x)$ on $\{a, b\}$, then :

(A) $\int_a^b f(x) dx \geq \int_a^b g(x) dx.$

(B) $\int_a^b f(x) dx \leq \int_a^b g(x) dx.$

(C) $\int_a^b f(x) dx = \int_a^b g(x) dx.$

(D) $\int_a^b f(x) dx = -\int_a^b g(x) dx.$

SECOND SEMESTER (CBCSS-UG) DEGREE EXAMINATION, APRIL 2021

Mathematics

MTS 2B 02—CALCULUS OF SINGLE VARIABLE – I

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Find two functions f and g such that $F = g \circ f$ if $F(x) = (x+2)^4$.

2. Let $f(x) = \begin{cases} -x+3 & \text{if } x < 2 \\ \sqrt{x-2}+1 & \text{if } x \geq 2 \end{cases}$

Find $\lim_{x \rightarrow 2} f(x)$ if it exists.

3. Find the values of x for which the function $f(x) = x^8 - 3x^4 + x + 4 + \frac{x+1}{(x+1)(x-2)}$ is continuous.

4. Find $\lim_{x \rightarrow 0} \frac{\tan x}{x}$.

5. Find the instantaneous rate of change of $f(x) = 2x^2 + 1$ at $x = 1$.

6. If $f(x) = 2x^3 - 4x$. Find $f'(-2)$ and $f'(0)$.

7. Find the extreme values $f(x) = 3x^4 - 4x^3 - 8$ on $[-1, 2]$.

8. Determine where the graph of $f(x) = x^3 - 6x$ is concave upward and where it is concave downward.

9. Find $\lim_{x \rightarrow -1} \frac{1}{x+1}$.

10. Find the horizontal asymptote of the graph of $f(x) = \frac{1}{x-1}$.

Turn over

11. Find $\int \frac{\sin t}{\cos^2 t} dt$
12. Find $\int \frac{1}{x \log x} dx$.
13. Given that $\int_2^2 f(x) dx = 3$ and $\int_0^2 f(x) dx = 2$, evaluate $\int_2^0 f(x) dx$.
14. Find the area of the region bounded by the graphs of $y = 2 - x^2$ and $y = -x$.
15. Find the volume of the solid obtained by revolving the region bounded by $y = x^3$, $y = 8$ and $x = 0$ about the y -axis.

(10 × 3 = 30 marks)

Section B

Answer at least five questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

16. Show that the function $f(x) = |x|$ is differentiable everywhere except at 0.
17. Show that if the function f is differentiable at a , then f is continuous at a .
18. Verify that the function $f(x) = x^2 + 1$ satisfies the hypothesis of the mean value theorem on $[0, 2]$ and find all values of c that satisfy the conclusion of the theorem.
19. Find the relation extrema if any of the function $h(t) = \frac{1}{3}t^3 - 2t^2 - 5t - 10$.
20. The velocity function of a car moving along a straight road is given by $v(t) = t - 20$, for $0 \leq t \leq 40$, where $r(t)$ is measured in feet per second and t in seconds. Show that at $t = 40$, the car will be in the same position as it was initially.
21. (a) State mean value theorem for integrals.
(b) Verify mean value theorem for $f(x) = x^2$ on $[1, 4]$.

22. (a) Use differentials to obtain an approximation of the arc length of the graph of $y = 2x^2 + x$ from P (1,3) to Q (1.1, 3.52).
- (b) Find the work done in lifting a 50 – lb sack of potatoes to a weight of 4 ft above the ground.
23. Find the length of the graph of $f(x) = \frac{1}{3}x^3 + \frac{1}{4x}$ on the interval [1, 3].

(5 × 6 = 30 marks)

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. (a) Find the slope and an equation of the tangent line to the graph $f(x) = x^2$ at the point (1, 1).
- (b) Suppose that the total cost in dollars incurred per week by a company in manufacturing x refrigerators is given by the total cost function $c(x) = -0.2x^2 + 200x + 9000$, $0 \leq x \leq 400$.
- (i) What is the cost incurred in manufacturing the 201 st refrigerator ?
- (ii) Find the rate of change of c with respect to x when $x = 200$.
25. A man has 100 ft of fencing to enclose a rectangular garden. Find the dimensions of the garden of largest area he can have if he uses all of the fencing.
26. (a) Estimate $\int_0^1 e^{-\sqrt{x}} dx$ using the property of definite integral.
- (b) Use the geometric interpretation of the integral to evaluate $\int_{-1}^2 |x-1| dx$ by making a sketch of f .
27. Find the area of the surface that is generated by revolving the portion of the curve $y = x^3$ between $x = 0$ and $x = 1$ about the x -axis.

(2 × 10 = 20 marks)